

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI (PILANI CAMPUS)
FIRST SEMESTER 2016 – 2017

Course: CE F214 Construction Materials
Component: Comprehensive Exam (Closed Book)

Date: 05-12-2016 (9:00 - 12:00 FN)
Max. Marks: 94

Part A

[11×2=22]

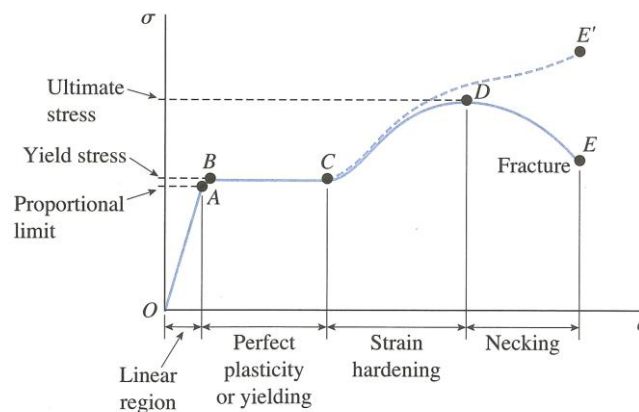
1. Although glass facade provides aesthetic value in the building, what is its disadvantage in terms of energy consumption?

Due to Infra red rays increase in the temperature and more energy consumption of air conditioning .

2. A building is constructed inside a national park where, animals and many birds are available. Decorative stone facade and glass facade are suggested to use in the building. Write your preference and justify the selection.

Stone façade. Glass has a tendency to reflect light. Also it makes an object look at greater distance than in reality. Thus when a bird flies towards the glass façade it assumes a bird flying from the opposite direction from a far off distance. It thus strikes the façade and gets killed due to the impact.

3. Draw stress-strain curves for a ductile material under uniaxial loading. Mark different zones



4. OPC 43 grade cement was used for pozzolanic activity test. Strength of 100% cement used mortar was observed 80% of the expected strength after 28 days of curing. Fly ash was used as a pozzolan. To evaluate reactivity of fly ash, mortar specimens were cast as per standard. Average compressive strength of fly ash blended mortar specimens (20 % level of replacement) was found as 36 MPa. Therefore, PAI of fly ash is.....

Strength of 100% cement mortar = $0.8 \times 43 = 34.4 \text{ Mpa}$

Strength of PPC = 36 MPa

*PAI = $\frac{\text{Strength of PPC}}{\text{Strength of OPC}} \times 100$
 $= \frac{36}{34.4} \times 100 = 104.65\%$*

5. A concrete mix was prepared and used to make 150×150×150 mm cube specimens and the same concrete is also used to make cylinder specimens as per ASTM standard. Average compressive strength of cube specimens after 28 days of controlled curing was observed as 32 MPa, then average compressive strength of cylindrical specimens is

Compressive Strength of cube=32MPa

$$\begin{aligned}\text{Compressive Strength of cylinder} &= 0.8 \times \text{Compressive Strength of cube} \\ &= 0.8 \times 32 \\ &= 25.6 \text{ MPa}\end{aligned}$$

6. Tensile testing of plain steel and HYSD steel was carried out. Yield point for plain steel was observed as 69% of the standard value whereas yield point of HYSD steel was observed as 88% of the standard value. What is the difference between yield stress of HYSD and plain steel.....192.7..... in MPa
7. In elongation and flakiness test after sieving process, the aggregates passing through 16 mm sieve and retained on 12.5 mm sieve are separated. Suitable slot is selected in test apparatus for flakiness test as specified in the Indian standard and similarly suitable slot is also selected in the length gauge for elongation test. What is the difference between selected slot length in elongation test to the selected width of slot in flakiness test.....17.1 mm.....
8. List different distresses/damages in flexible pavement

Different distresses in flexible pavement are:

Bleeding

Segregation

Polishing

Weathering

Aging

Thermal Cracks

Rutting

Shoving

Stripping

9. List different ingredients of oil paints

Base

Vehicle

Pigments

Solvents

Dryers

10. Write two commonly used activators in geopolymer concrete: 1)..... and 2).....

1)NaOH+Na₂SiO₃+Distilled Water

2)KOH+K₂SiO₃+Distilled Water

11. Average compressive strength of first class bricks was observed as 79% of the minimum value prescribed in the Indian standard. Moreover average compressive strength of second class bricks was also determined as 90% of the minimum value prescribed in the Indian standard. What is ratio of strength of second class bricks to the strength of first class bricks in MPa

Average strength of first class brick=10MPa

*Average strength of **given** first class brick=0.79*10=7.9MPa*

Average strength of second class brick=7MPa

*Average strength of **given** second class brick=0.9*7=6.3MPa*

Ratio=0.797

Part B

[72]

1. Write a brief note on complete processing of timber from cutting stage to preservation stages (10)

Main Processes:

1) Felling of Trees : *To be done when sap movement minimum hence not in spring*

Tree should be 50-100 yrs old

In hilly areas-Mid summers to avoid rains

In plane areas-Mid winters to avoid excessive moisture loss

2) Seasoning of timber: *Removing moisture from the timber. Methods of seasoning:*

i] Boiling

ii]Electrical Seasoning

iii]Kiln Seasoning

iv]Water Seasoning

v]Chemical Seasoning

3) Conversion of Timber: *Process of giving desired shape and size to timber section. Types of sawing:*

i]Ordinary Sawing

ii]Quarter Sawing

iii]Tangential Sawing

iv]Radial Sawing

4) Preservation of Timber: *Done to increase the life of timber and protect it from moisture, insect and termite attacks. Methods of Preservation are:*

i]AS-CU Treatment (1 part Arsenic Pentaoxide, 3 Parts CuSO₄, 4 Parts Sodium/Potassium Dichromate in H₂O.

ii] Application of chemical salts

iii] Coal Tar

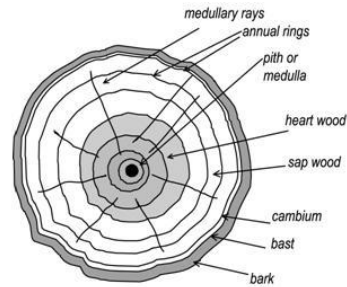
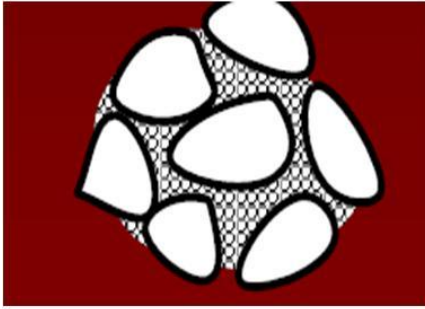
iv]Application of Creosote Oil

v]Application of Oil Paints

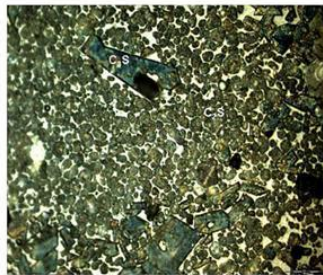
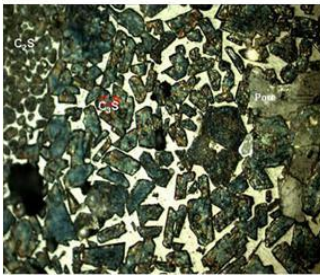
2. Draw neat sketches for the followings concepts (**only drawing; no explanation**) (20)

1. Optical microscopic structure of Alite and Belite

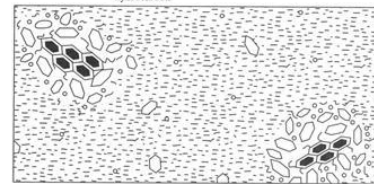
The large irregular crystals are grains of alite (C₃S). The more rounded smaller crystals are belite (C₂S), while the white groundmass consists of the flux phases



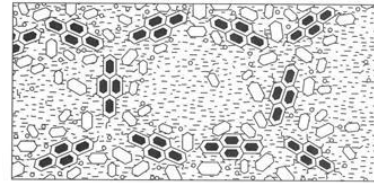
The tree trunk showing growth rings



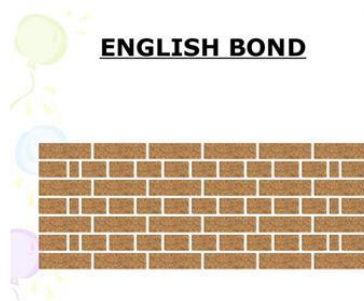
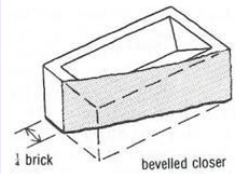
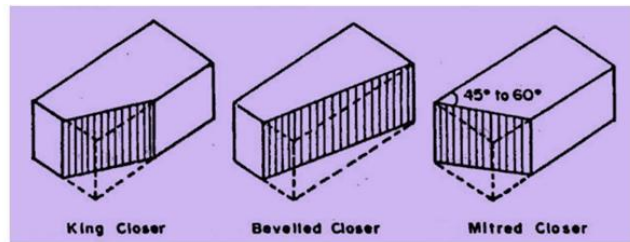
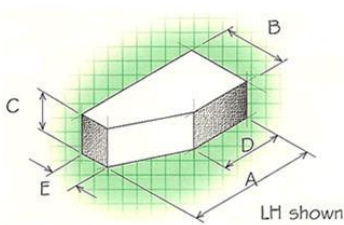
- Asphaltenes
- Aromatic/naphthenic hydrocarbons
- High-molecular-weight aromatic hydrocarbon
- Naphthenic/aliphatic hydrocarbons
- Low-molecular-weight aromatic hydrocarbon
- Saturated hydrocarbons



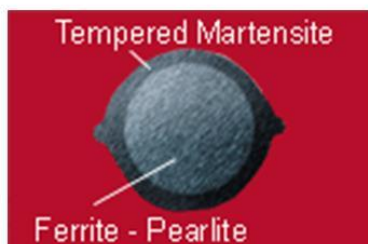
(a)



(b)

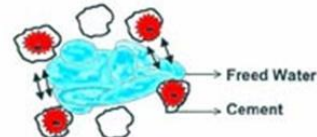


ENGLISH BOND

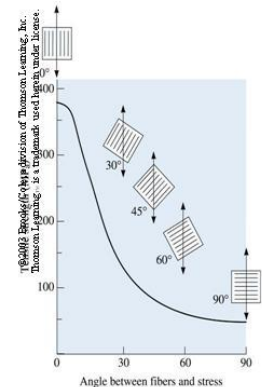
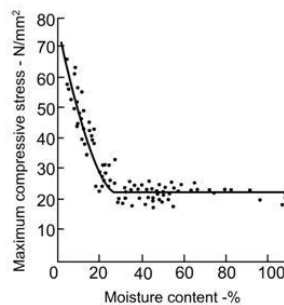


1. Water Reducer coats the cement grains

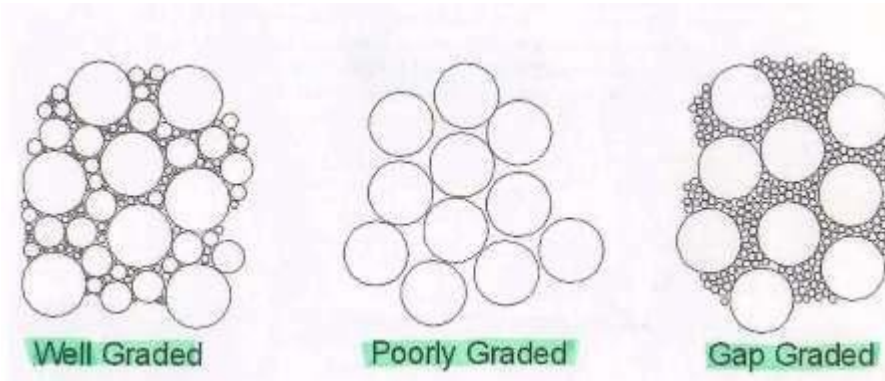
2. Water Reducer imparts negative charge to cement grains



3. Like charges repel cement grains away from each other

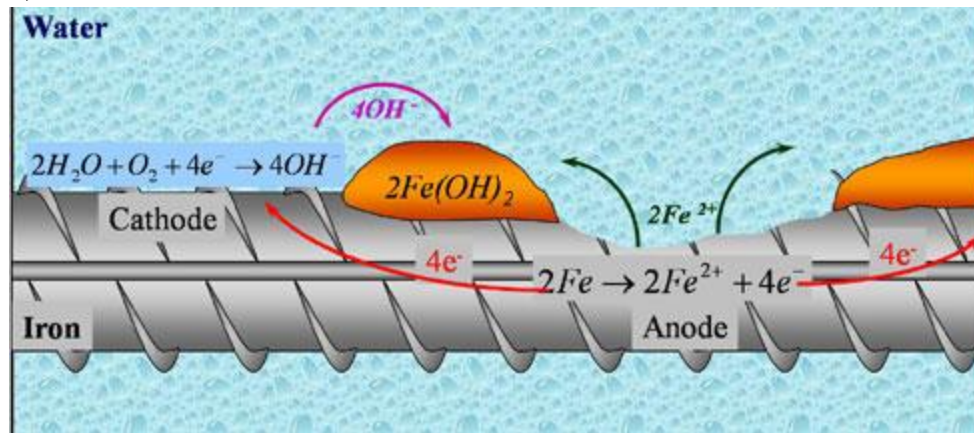


Gap gradation in aggregate



3. Explain the following

Corrosion mechanism of steel reinforcement including anodic and cathode reactions with a neat sketch (3+2)



Anode: $\frac{1}{2} O_2 + H_2O + 2e^- \rightarrow 2OH^-$

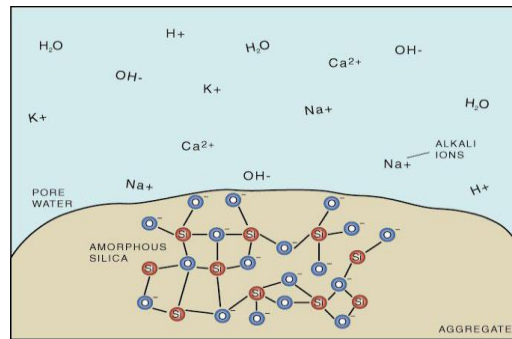
Cathode: $Fe \rightarrow Fe^{2+} + 2e^-$

List different types of reinforcements (3)

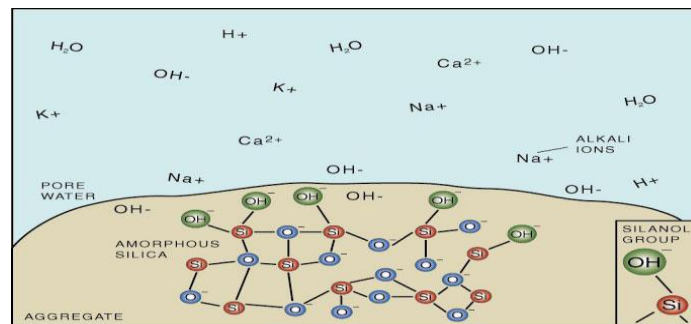
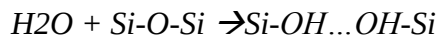
- *Uncoated steel*
 - Mild steel ribbed bars
 - High Yield Strength Deformed (HYSD) bars
 - Cold-twisted deformed (CTD) bars (TORsteel)
 - Thermomechanically treated (TMT) or Quenched and Self-Tempered (QST) steel
 - Stainless steel
 - Prestressing steel
 -
- *Coated steel*
 - Epoxy coated steel
 - Galvanized steel

4. Sketch six major stages of alkali silica reaction. Write short note on each stage. (6+2)

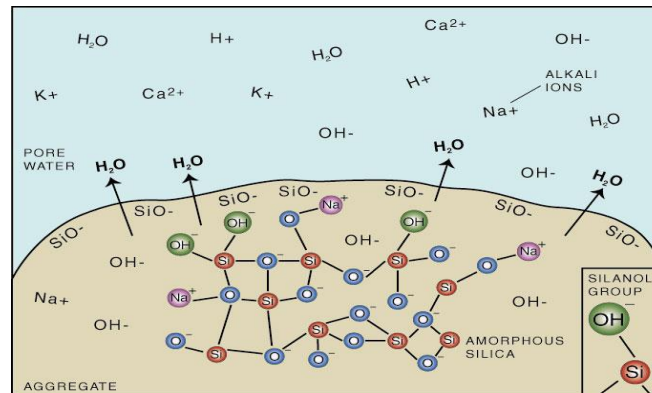
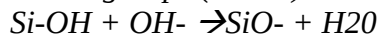
1. Aggregate in solution, pre-ASR damage



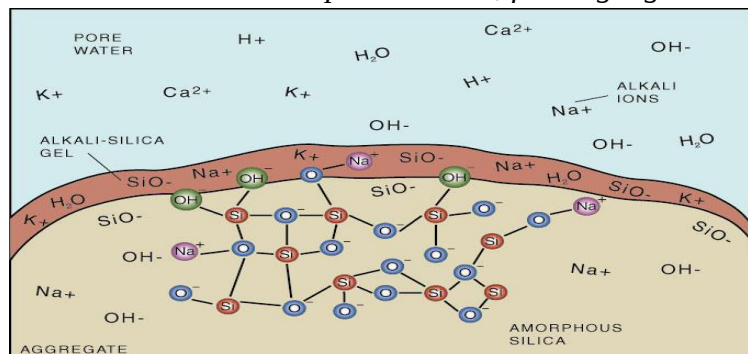
2) Surface of Agg attacked by OH^-



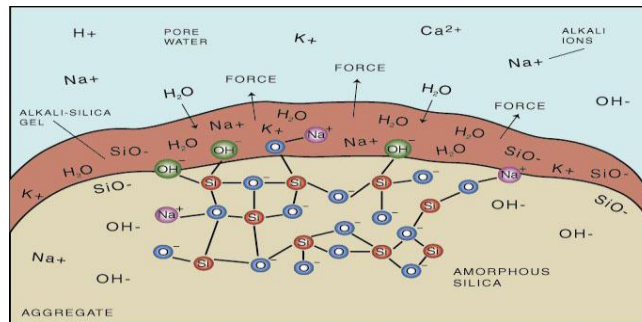
3) Silanol groups ($Si-OH$) on surface are broken down by OH^- into SiO^- molecules



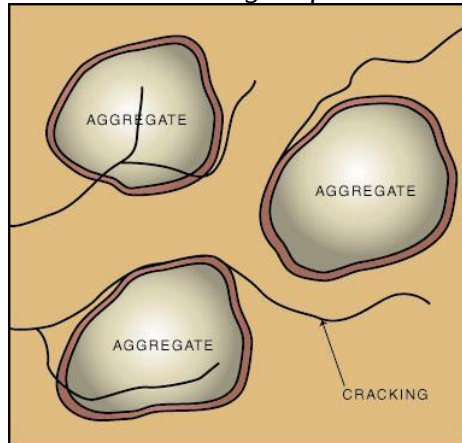
4) Released SiO^- molecules attract alkali cations in pore solution, forming a gel around the aggregate



5) Alkali-silica gel takes in water, expanding and exerting a force against surrounding concrete.



6) When the expansive pressure exceeds the tensile strength of the concrete, the Concrete cracks



5.Design the following mixes using details given below (**Tables are given in the next page**)

(a) M25 mix using step-by step procedure as per IS 10262-2009 (7)

Step 1:-

Target Mean Strength of Concrete (ft)	=	$f_{ck} + (1.65 S)$
f_{ck}	=	<u>25</u> N/mm ²
Standard Deviation (S)	=	<u>4</u>
(Read from Table - 1)		
ft	=	<u>$25 + (1.65 \times 4)$</u>
	=	<u>31.6</u> N/mm ²

Step 2:-

(i) W / C ratio (Assumed based on experience)	=	<u>0.4</u>
(ii) Limiting or Maximum W / C ratio	=	<u>0.5</u> (for Severe condition of Exposure)
(Read from Table - 5 of IS:456 - 2000)		
Compare & Fix W / C ratio	=	<u>0.4</u> [0.4 < 0.5; Hence OK]

Step 3 & 4:-

Water Content	=	<u>186</u> litres / m ³ of Concrete
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(Read from Table - 2)

(for 25 to 50 mm Slump)

$$\begin{aligned} \text{Increase in Water Content} &= \frac{186 \times 1.03}{\text{litres / m}^3 \text{ of Concrete}} \\ \text{(3\% increase for every 25 mm extra slump)} & \text{(3\% increase for 75 mm slump)} \\ &= \underline{191.6} \text{ litres / m}^3 \text{ of Concrete} \end{aligned}$$

$$\begin{aligned} \text{Decrease in Water Content} &= \frac{191.6 \times 0.71}{\text{litres / m}^3 \text{ of Concrete}} \\ \text{(20\% \& above due to addition of Superplasticizer)} & \text{(29\% decrease based on trial)} \\ &= \underline{136.022} \text{ litres / m}^3 \text{ of Concrete} \end{aligned}$$

Step 5:-

$$\text{W / C ratio} = \underline{0.4}$$

$$\text{(1)Cement Content} = \frac{\text{Water Content}}{\text{W / C ratio}}$$

$$= \frac{136.022}{0.4}$$

$$= \underline{340.05} \text{ Kg}$$

$$\begin{aligned} \text{(2)Minimum Cement Content} &= \underline{300} \text{ Kg (for Severe Exposure Condition)} \\ \text{(Read from Table - 5 of IS: 456 - 2000)} & \end{aligned}$$

$$\begin{aligned} \text{(3)Maximum Cement Content} &= \underline{450} \text{ Kg} \\ \text{(As per Clause 8.2.4.2 of IS: 456 - 2000)} & \end{aligned}$$

$$\text{Fix Cement Content (C)} = \underline{350} \text{ Kg} \quad (300 < 350 < 450)$$

Step 6:-

$$\begin{aligned} \text{Volume of Coarse Aggregate} & \\ \text{per unit volume of Total Aggregate} &= \underline{0.62} \text{ (for W/C ratio of 0.5)} \\ \text{(Read from Table - 3 against 20 mm agg.} & \\ \text{\& Zone - II Fine aggregate)} & \end{aligned}$$

Step 7:-

$$(a) \text{W/C ratio adopted} = \underline{0.4}$$

For every 0.05 increase / decrease in W/C ratio from 0.5, reduce / add 0.01

$$\text{Therefore,} \quad \underline{\text{Add 0.02}}$$

$$\begin{aligned} \text{Now, the Volume of Coarse Aggregate} \\ \text{per unit volume of Total Aggregate} \end{aligned} = \underline{0.64}$$

$$\begin{aligned} (b) \text{For Pumpable Concrete, reduce by 10\%} \\ \\ \end{aligned} = \underline{0.64 \times 0.9}$$
$$= \underline{0.576}$$

$$\begin{aligned} \text{Corrected Volume of Coarse Aggregate} \\ \text{per unit volume of Total Aggregate} \end{aligned} = \underline{0.576}$$

$$\begin{aligned} \text{Hence Volume of Fine Aggregate} \\ \text{per unit volume of Total Aggregate} \end{aligned} = \underline{(1 - 0.576)}$$
$$= \underline{0.424}$$

Step 8:-

$$(1) \text{ Volume of Concrete} = \underline{1} \text{ m}^3$$

$$\begin{aligned} (2) \text{ Volume of Cement} &= \frac{\text{Mass of Cement}}{\text{Sp. Gr. of Cement} \times \text{Density of Water}} \\ &= \frac{340.05}{\underline{3.15} \times \underline{1000}} \\ &= \underline{0.108} \text{ m}^3 \end{aligned}$$

$$\begin{aligned} (3) \text{ Volume of Water} &= \frac{\text{Mass of Water}}{\text{Sp. Gr. of Water} \times \text{Density of Water}} \end{aligned}$$

$$= \frac{136.02}{\frac{1}{1} \times 1000}$$

$$= 0.136 \text{ m}^3$$

(4) Volume of Chemical admixture
(Superplasticizer @ 2% by mass of
Cementitious material)

$$= \frac{\text{Mass of admixture}}{\text{Sp. Gr. of admixture} \times \text{Density of Water}}$$

$$= \frac{(340.05 \times 0.02)}{\frac{1.145}{1} \times 1000}$$

$$= 0.006 \text{ m}^3$$

(5) Combined Vol. of Coarse & Fine
Aggregate

$$= [(1) - \{ (2) + (3) + (4) \}]$$

$$= \{1 - (0.108 + 0.136 + 0.006)\}$$

$$= 0.75 \text{ m}^3$$

(6) Mass of Coarse Aggregate

$$= 0.75 \times 0.576 \times 2.74 \times 1000$$

$$= (5) \times \text{Vol. of C.A.} \times \text{Sp.Gr. of C.A.} \times \text{Density of Water}$$

$$= 1183.77 \text{ Kg}$$

(7) Mass of Fine
Aggregate

$$= 0.75 \times 0.424 \times 2.76 \times 1000$$

$$= (5) \times \text{Vol. of F.A.} \times \text{Sp.Gr. of F.A.} \times \text{Density of Water}$$

$$= 877.68 \text{ Kg}$$

Step 9:- Mix Proportion

Water	Cement	Fine Aggregate	Coarse Aggregate	
(litres)	(Kg)	(Kg)	(Kg)	
136	340.05	877.68	1183.77	(per m ³ of Concrete)

0.4 : 1 : 2.58 : 3.481 (Ratio)

Result:-

**The Concrete Mix Proportion is 1 : 2.58 : 3.48
with a W / C ratio of 0.4 & suitable dosage of Admixture**

(c) List your observations on salient features in both mixes (2)

For both the mixes the mix proportion changes only when we assume different w/c ratio. Other factors such as size and zone of fine aggregate are considered same for both the cases and hence assuming same w/c ratio will give no change in the final ratio.

5. List different types of concrete (2)

High Strength concrete

High Performance

Self Compacting

Fiber Reinforced

Lightweight

Ultra lightweight

Shotcrete

Porous/Permeable

Roller Compacted

Heat resistant

Recycled Aggregate Concrete

Polymer Modified

Explain characteristics, ingredients and application for any four types of concrete (8)

1) ***Heat Resistant Concrete:***

Characteristics-Concrete can be exposed to temperature as high as 1000C without causing thermal cracks or any other damage.

Ingredients-Made with high alumina cement which contains 40% Alumina and 40% Lime

Applications- Used in structures exposed to very high temperature like nuclear reactors, blast furnace etc.

2) ***Fiber Reinforced Concrete:***

Ingredients: Fibers such as steel fiber and carbon fiber are crushed in small pieces for better mixing and then added.

Characteristics – Fibers increase post cracking strength and stitch the cracks in perpendicular direction. Helps in taking extra load.

Applications-Used in constructing bridges

3) ***Self Compacting Concrete:***

Ingredients: Large amount of water. Super plasticizers to increase the workability and fine aggregate.

Characteristics: Should be resistant to segregation, have flowing and passing ability.

Applications: Used when pumping of the concrete is done and when spacing between the concrete is very less making manual application very difficult.

4) **Shotcrete :**

Characteristics: Should have moderate fluidity and workability. Concrete sets uniformly due to high pressure with which it emitted.

Ingredients: OPC with less water to keep workability and fluidity in check

Application: Majorly used in tunneling.

Sl No. (1)	Grade of Concrete (2)	Assumed Standard Deviation N/mm ² (3)
i) ii)	M 10 M 15	3.5
iii) iv)	M 20 M 25	
v) vi) vii) viii) ix) x)	M 30 M 35 M 40 M 45 M 50 M 55	4.0
		5.0

Table 2 Maximum Water Content per Cubic Metre of Concrete for Nominal Maximum Size of Aggregate
(Clauses 4.2, A-5 and B-5)

Sl No.	Nominal Maximum Size of Aggregate mm	Maximum Water Content ¹⁾ kg
(1)	(2)	(3)
i)	10	208
ii)	20	186
iii)	40	165

Table 3 Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate
(Clauses 4.4, A-7 and B-7)

Sl No.	Nominal Maximum Size of Aggregate mm	Volume of Coarse Aggregate ¹⁾ per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
		Zone IV	Zone III	Zone II	Zone I
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	0.50	0.48	0.46	0.44
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.75	0.73	0.71	0.69